

Small object detection using integrated models

VisDrone and KITTI Benchmark Results

Abstract

Small object detection in aerial and road imagery remains challenging due to limited resolution, cluttered backgrounds, and spatial degradation through processing layers. To improve the performance for small objects, we propose an integrating approach combining YOLO11n, Slicing Aided Hyper Inference (SAHI) and SAM 3 as a post-processing segmentation step. Our study evaluates five detection pipelines combining YOLO11n, Slicing Aided Hyper Inference (SAHI), and Meta Segment Anything Model 3 (SAM 3) on VisDrone2019-DET small object dataset. On VisDrone, the full YOLO11n+SAHI+SAM 3 pipeline achieves the best mAP@50 (0.409), F1 (0.478), and precision (0.569), demonstrating that segmentation-guided refinement meaningfully improves detection of small aerial objects.

1. Introduction

Small Object Detection is important in aerial and road imagery. It contributes in various applications such as precision agriculture, disaster response, reading traffic lights and road signage among many other things. Detecting small objects is not an easy task as they can often be missed, detected wrongly, or even mislabeled upon detection. There are multiple causes for these failures, all of which compound on one another. Beginning from the limited information of the small object from the input image or video frame, followed by the spatial degradation of the input as it passes through multiple layers of processing such as removal of complex background objects or downsampling [11, 12].

You Only Look Once (YOLO), a type of Convolution Neural Network (CNN), is one of the most influential methods for small object detection due to its speed and efficiency [13, 8]. While YOLO methods are balanced they have their limitations in small object detection in high resolution images, particularly in complex backgrounds [7]. This might be due to the post-processing techniques such as Non-Maximum Suppression (NMS) which makes them slower and introduces hyperparameters that can result in instability [2].

To address the shortcomings of YOLO, the Slicing Aided Hyper Inference (SAHI) method is implemented. SAHI slices large images into smaller overlapping tiles. This breakdown allows the detector to direct focus on local areas without giving up the higher resolution. This localized focus and analysis increases detection of small objects that are conventionally missed through global image analysis [14]. SAHI has been implemented in various applications such as aerial and road images [9, 15].

To further improve the results of YOLO, we implement another segmentation model known as Segment Anything Model (SAM) [6]. This is another deep learning model used in object segmentation. Accurate segmentation which reflects the distinctive features of each object can affect how well they are classified by the model. The SAM does this by extracting low and high level features from the images such as position, shape or any other defining feature [5, 10].

The integration of a segmentation algorithm after detection as a post-processing step is understudied and is a gap in the literature this work aims to fill. Pervious work only address the integration of YOLOv8 and SAHI [3]. This work tests the integration of YOLO, SAHI and SAM, in aerial and road imagery. The primary contributions of this work are:

1. Evaluate performance of YOLO11n independently.
2. Evaluate performance of YOLO11n integrated with SAHI.
3. Evaluate performance of SAM3 independently.
4. Evaluate performance of YOLO11n integrated with SAM3.

5. Evaluate performance of YOLO11n integrated with SAHI and SAM3.

All the above are tested on the VisDrone2019-DET [16] and the KITTI [1] datasets.

2. Methods

2.1. Datasets

2.1.1 VisDrone2019-DET

This dataset is part of the Vision initiative. It comprises high-resolution aerial imagery collected using drones over multiple urban areas in China. The dataset comprises 6,471 images for training, 548 for validation, and 1,610 for testing, with an average resolution of 1920×1080 pixels [16]. It features 10 predefined object classes, including pedestrian, car, bicycle, van, truck, tricycle, awning-tricycle, bus, motor, and others. Each image is labeled with bounding boxes to indicate object locations, class identifiers, and visibility indicators. A standard input of this dataset is represented in Figure 1.

2.1.2 KITTI

The Kitti dataset is collected by a vehicle equipped with dash camera and other sensors for autonomous driving testing and benchmarking [1]. The Kitti dataset consists of 5,985 images for training images and 1,496 images for validation. It features seven labeled classes such as cars, van, tram, trucks, pedestrian, person sitting and cyclist. A standard input of this dataset is represented in Figure ??.

2.2. Models

2.2.1 YOLO11n

YOLO11n, introduced by Ultralytics, builds upon its predecessors, particularly YOLOv8, by enhancing feature extraction capabilities, optimized speed and greater accuracy all the while supporting more diverse vision tasks [4]. In this study we will use the YOLO11n model for detection. The model was pretrained on the COCO dataset and fine tuned for both datasets independently for 50 training epochs and a consistent image size of 640px.

2.2.2 SAHI

Slicing Aided Hyper Inference (SAHI) slices large images into smaller overlapping tiles. SAHI consists of three main steps: slicing, detection and merging. For slicing, a tile size of 256x256 with 0.5 overlap ratio were selected. For detection, a threshold confidence of 12 tiles was selected. For merging, a GREEDYNMM post-processing approach was taken with an IoU of 0.4 and deduplication of 0.5.

2.2.3 SAM 3

Segment Anything Model 3 (SAM 3) is a foundation model for Promptable Concept Segmentation (PCS). SAM 3 is capable of detecting, segmenting, and tracking all instances of a visual concept specified by text prompts, image exemplars, or both. SAM 3 consists of a detector and tracker that share a Perception Encoder (PE) vision backbone. This decoupled design avoids task conflicts while enabling both image-level detection and video-level tracking. The pretrained benchmark model of this was used in this study.

2.3. Experimental Parameters

All experiments were performed on a NVIDIA RTX 4060 Laptop GPU (8 GB VRAM).

2.4. Experimental Pipelines

This study explores five different detection pipelines. These are listed below:

1. **YOLO11n**: This pipeline employs the YOLO11n model fine-tuned to each dataset.
2. **YOLO11n+SAHI**: This pipeline employs the fine-tuned YOLO11n model with SAHI slicing.

3. **SAM 3**: This pipeline employs SAM 3 text prompts per class.
4. **SAM 3+YOLO11n**: This pipeline employs YOLO11n detection and SAM 3 masks and text proposals.
5. **YOLO11n+SAHI+SAM 3**: This pipeline employs the fine-tuned YOLO11n with SAHI slicing and the masking employed by the SAM 3 model.

3. Results and Discussion

We assess results based on precision, recall, F1, mean Average Precision (mAP) and frames per second (FPS).

Precision highlights the proportion of correctly classified positive instances out of all instances predicted as positive generated by the model, while recall shows the proportion of actual positive instances that the model successfully detects. F1-score is the harmonic mean of precision and recall, this is a balanced and comprehensive metric.

The mean Average Precision (mAP) is the mean of multiple Average Precision (AP) scores across different classes or thresholds computed for each class. The mAP at an IoU threshold of 0.5 (mAP@0.5), and mAP averaged across IoU thresholds from 0.5 to 0.95 (mAP@0.5:0.95).

After testing the five pipelines on the VisDrone and KITTI datasets. The performance comparison on the VisDrone dataset is presented in Table 1. The detection masks and overlays of the pipeline are presented in Figure 2.

Table 1. Performance Comparison on VisDrone Dataset

Pipeline	Precision	Recall	F1	mAP@50	mAP@50:95	FPS
YOLO11n	0.487	0.388	0.462	0.274	0.169	75.5
YOLO11n+SAHI	0.161	0.501	0.244	0.370	0.226	1.09
SAM 3	0.350	0.234	0.280	0.301	0.114	0.27
SAM 3+YOLO11n	0.502	0.392	0.440	0.365	0.218	0.06
YOLO11n+SAHI+SAM 3	0.569	0.469	0.478	0.409	0.255	0.02

Best values are in **BOLD**

From the results, we can see that YOLO11n+SAHI+SAM 3 pipeline leads in precision, F1, mAP@50 and mAP@50:95. However it is might be ideal if you need fast processing as it processes at 00.02 FPS (~44 s per image). YOLO is the fastest in detection at 75.5 FPS but lowest mAP@50 (0.274).

YOLO11n+SAHI has high recall (0.501) but low precision (0.161), it also had 47261 duplicate detections along with 196106 false positives. SAM 3 has the lowest recall (0.234) and mAP@50 (0.114).

The performance comparison on the KITTI dataset is presented in Table 2. The detection masks and overlays of the pipeline are presented in Figure ??.

Table 2. Performance Comparison on KITTI Dataset

Pipeline	Precision	Recall	F1	mAP@50	mAP@50:95	FPS
YOLO11n	0.781	0.920	0.844	0.834	0.597	71.2
YOLO11n+SAHI	0.563	0.681	0.617	0.689	0.407	2.23
SAM 3	0.084	0.679	0.149	0.802	0.425	1.15
SAM 3+YOLO11n	0.506	0.861	0.637	0.528	0.379	0.04
YOLO11n+SAHI+SAM 3	0.495	0.868	0.631	0.530	0.381	0.16

Best values are in **BOLD**

From the results, we can see that YOLO11n pipeline leads in precision, recall, F1, mAP@50, mAP@50:95 and FPS. We can see that the introduction of SAHI YOLO11n+SAHI drops mAP@50 from 0.834 to 0.689. SAM 3 alone had the lowest precision but retained an F1 score of 0.149 because of the high recall of 0.679.

We can see from these results that the performance gains seen in the VisDrone dataset do not carry over to the KITTI dataset. This can be due to masking limitations or hardware capabilities in this study.

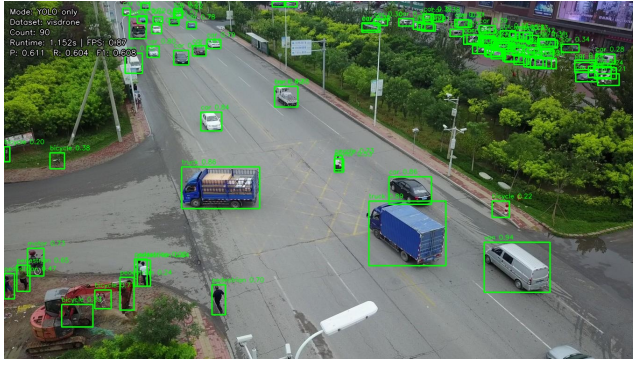
4. Conclusion

This study benchmarked five detection pipelines combining YOLO11n, SAHI, and SAM 3 on the VisDrone2019-DET dataset for small object detection in aerial imagery. The full YOLO11n+SAHI+SAM 3 pipeline achieved the best overall

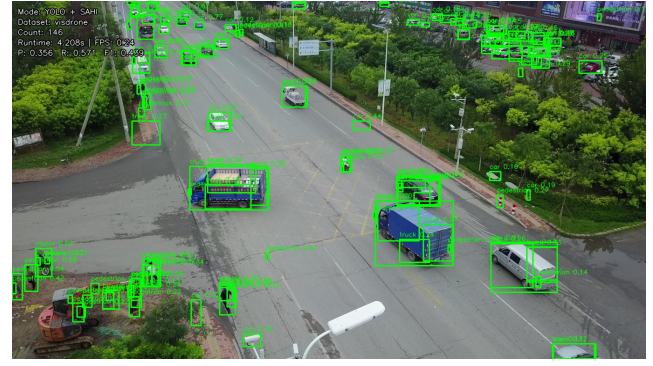
accuracy with mAP@50 of 0.409 and F1 of 0.478, demonstrating that post-processing segmentation via SAM 3 meaningfully refines detections produced by YOLO11n and SAHI. Among the simpler pipelines, YOLO11n alone remained the fastest at 75.5 FPS, making it the practical choice for real-time applications despite its lower accuracy. The key limitation of the full pipeline is its computational cost at approximately 0.02 FPS, which currently restricts deployment to offline or low-latency-tolerant scenarios. Future work will focus on accelerating the integrated pipeline through lightweight segmentation alternatives to SAM 3 and more efficient merging strategies, with the goal of closing the gap between accuracy and real-time performance.



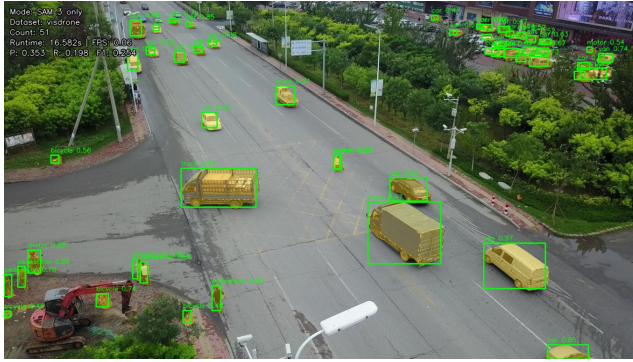
Figure 1. VisDrone Input Image



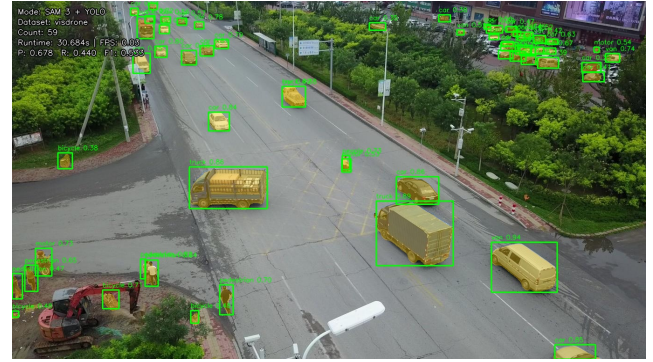
(a) YOLO11n



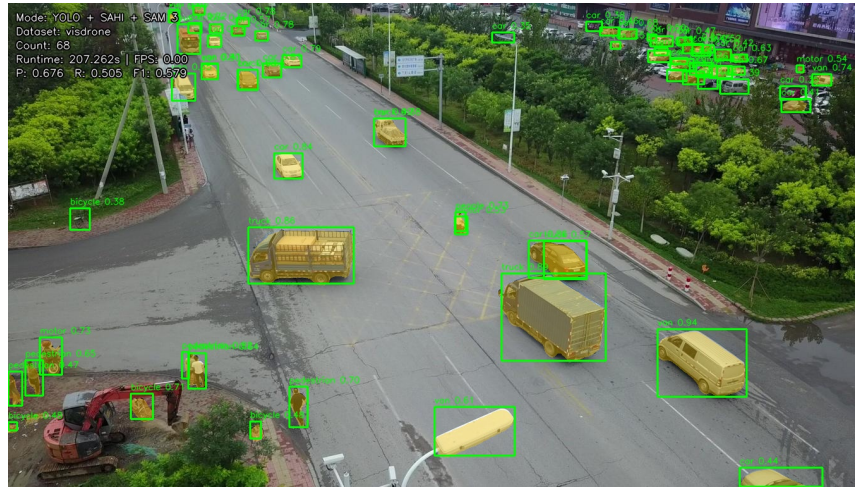
(b) YOLO11n+SAHI



(c) SAM 3



(d) SAM 3+YOLO11n



(e) YOLO11n+SAHI+SAM 3

Figure 2. VisDrone Detection and Mask Overlays for all five pipelines

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